

PROJECT REQUIREMENTS

Project Name: College Internship Ecosystem

Prepared By: Duddukuri Avinash

Problem Statement & Core Product Vision:

Students often struggle to find relevant internship opportunities, understand which opportunities fit their skills and eligibility, track applications, and learn from seniors who have already completed internships. At the college level, internship information may also be scattered across faculty messages, student groups, placement communications, and external websites. This makes it difficult for students to discover trustworthy opportunities and for faculty to monitor internship progress.

The proposed platform is a college-focused internship ecosystem that brings opportunities, student profiles, applications, document verification, senior experiences, and internship progress into one system. The core philosophy is to make relevant opportunities easier to discover while reducing repetitive work for faculty through automation. Important documents and internship records remain subject to human verification so that the platform maintains trust and accuracy.

Core Functional Requirements:

A. Student Registration & Profile

- **College Authentication:** Students register using Roll Number and college email.
- **Profile Setup:** Branch, year, section, skills, programming languages, tools, certifications and career preferences.
- **Resume:** Students can upload and maintain their latest resume.
- **Profile Completion:** The system automatically calculates profile completion and highlights missing information.

B. Internship Discovery & Personalized Recommendations

- **Central Internship Listing:** Students can browse college opportunities, self-found opportunities and senior referrals.
- **Search & Filters:** Company, role, branch, year, skills, location, mode, duration, stipend and deadline.
- **Eligibility Matching:** Automatically identifies opportunities suitable for the student's branch, year and stated skills.
- **Match Score:** Shows how well an internship matches the student's profile without guaranteeing selection.
- **Save & Apply:** Students can save opportunities and open the official application link.

C. Student Dashboard & Application Tracking

- **Personal Dashboard:** Recommended internships, deadlines, profile status, readiness and recent notifications.
- **Application Tracker:** Saved, Applied, Interview, Selected, Rejected and Withdrawn.
- **Deadline Alerts:** Automatic reminders for upcoming internship application deadlines.
- **Action Suggestions:** The system recommends useful next actions such as completing a profile, applying to matching internships or improving a skill.

PROJECT REQUIREMENTS — CONTINUED

D. Internship Progress & Document Workflow

- **Offer Letter Upload:** Students upload their offer letter after receiving an internship.
- **Human Verification:** Authorized faculty review the submitted document and choose Verify, Request More Information or Reject.
- **Additional Proof:** When information is missing, students can upload additional documents requested by faculty.
- **Completion Certificate:** Students upload the certificate after completing the internship.
- **Verification History:** Verification status, reviewer and important actions are recorded for accountability.

E. Learn From Seniors

- **Verified Experiences:** Students can view internship experiences submitted by seniors after moderation.
- **Experience Details:** How the internship was obtained, role, duration, skills used, project, interview/application experience, challenges and learning.
- **Filtering:** Experiences can be filtered by role, company, skills, branch and internship type.
- **Experience Moderation:** Faculty can approve, reject or request changes before an experience becomes visible as verified.

F. Faculty Portal

- **Faculty Registration:** Faculty register using Employee ID and official college email. College-email verification is used for account validation.
- **Faculty Dashboard:** Students, active internships, pending verification, completed internships and students needing attention.
- **Post College Opportunities:** Faculty can publish internship opportunities available through the college.
- **Automatic Eligibility:** The system identifies students who meet the opportunity's branch, year and skill requirements.
- **Targeted Notifications:** Faculty can notify eligible students instead of manually contacting students one by one.
- **Verification Center:** Central queue for offer letters, certificates, self-found internships and experiences.
- **Student Monitoring:** Faculty can identify students with no applications, incomplete profiles or internships requiring attention.
- **Announcements:** Faculty can send internship-related announcements to departments, years, sections or selected students.

G. Admin & Platform Management

- **Student Management:** View and manage student accounts and overall internship status.
- **Faculty Management:** View active faculty accounts and manage access/permissions when required. Faculty are not manually added by Admin.
- **Internship Management:** Monitor all internship listings and handle duplicate, inappropriate or reported content.
- **Moderation:** Manage reports and content issues.
- **Academic Settings:** Manage departments, academic years, sections and other college-level configuration.
- **Analytics:** College-wide internship statistics and skill-gap insights.
- **Audit Logs:** Record important administrative and verification actions.

TECHNOLOGY & PRODUCT STRATEGY

H. Automation & Intelligent Features

- **Automatic Eligibility:** Matches internship requirements with student academic details and skills.
- **Personalized Recommendations:** Ranks relevant internships based on profile information and preferences.
- **Deadline Notifications:** Automatically reminds students about important deadlines.
- **Duplicate Detection:** Flags potentially duplicate internship postings using company, role, URL and other details.
- **Skill-Gap Analysis:** Identifies commonly missing skills by comparing student profiles with internship requirements.
- **AI Assistance:** AI may assist with resume/readiness analysis and information extraction, but final document verification remains a human decision.

I. Security & Privacy Requirements

- Passwords must be securely hashed and never stored as plain text.
- Role-based access control for Student, Faculty and Admin.
- Resumes, offer letters and certificates must only be accessible to authorized users.
- File type and file-size validation for uploaded documents.
- Verification and administrative actions should have an audit trail.
- Students should not be shown private information belonging to other students.

Technology Stack

- **Frontend:** React.js + Vite — responsive student, faculty and admin interfaces.
- **UI:** HTML5, CSS3, JavaScript and a responsive component/UI library such as Tailwind CSS.
- **Backend:** Python + FastAPI — REST APIs, authentication, business logic and file-processing workflows.
- **Database:** MySQL — students, faculty, internships, applications, verification records, notifications and related data.
- **Authentication:** JWT-based authentication with role-based authorization; college-email verification for account registration.
- **File Storage:** Secure object/file storage for resumes, offer letters and certificates, with database references to uploaded files.
- **AI Layer:** Python-based AI/LLM integration for resume analysis, skill-gap analysis and recommendation assistance where required.
- **Version Control:** Git + GitHub.
- **Deployment:** Frontend and backend can be deployed separately using suitable cloud hosting; database hosted securely.

Product Model & Expected Impact

College-Focused Platform: The first version is designed for a single college rather than a multi-college marketplace. This keeps the data model, eligibility rules and workflows manageable while allowing the system to expand later.

Main Value: Students get one trusted place to discover relevant internships, track applications, learn from seniors and maintain verified internship records. Faculty get a centralized workflow for opportunities, student monitoring and document verification. Administrators get college-level control, analytics and accountability.

Key Principle: Automation should reduce repetitive work, but important verification decisions—especially offer letters, completion certificates and internship evidence—remain with authorized faculty members.

Expected Outcome: Improve internship discovery and participation, reduce scattered internship information, provide trustworthy student internship records, and create a reusable knowledge base from senior experiences.

Application Screenshots

College Internship Ecosystem (InternHub) - Student, Faculty and Admin portals

Figure 1: Student Portal - Dashboard

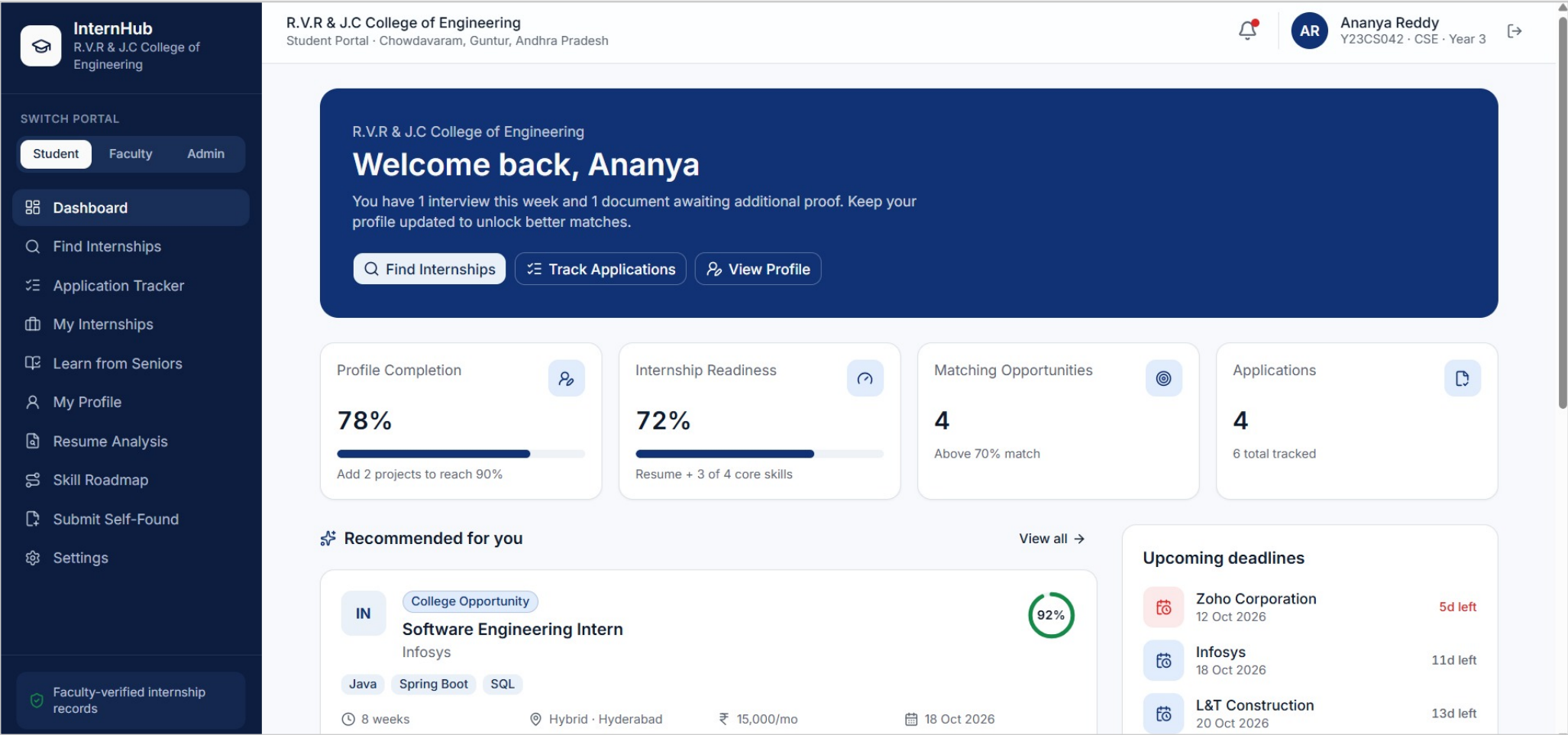


Figure 2: Faculty Portal - Dashboard

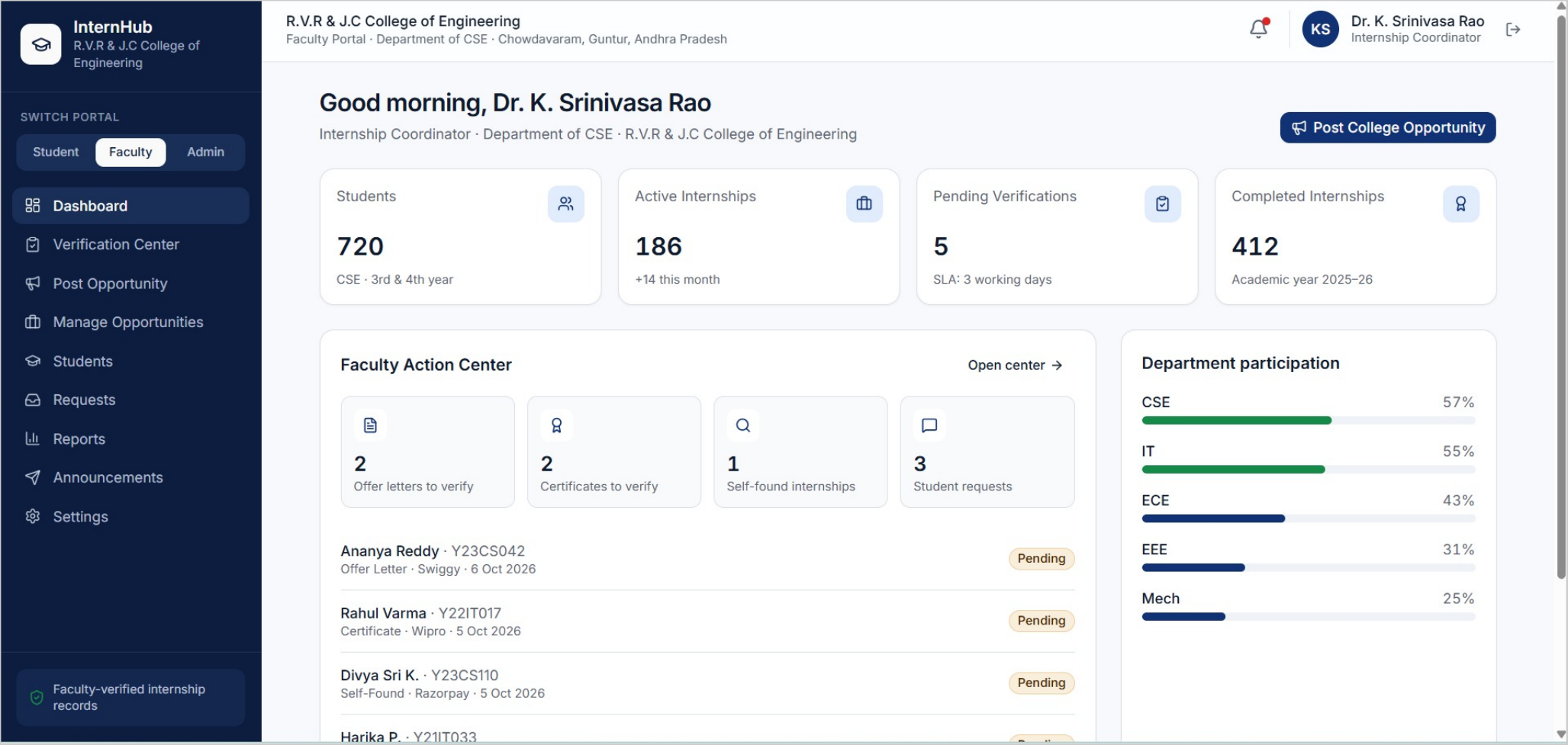


Figure 5: Faculty Portal - Post College Opportunity

InternHub

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SWITCH PORTAL

StudentFacultyAdmin

Dashboard

Verification Center

Post Opportunity

Manage Opportunities

Students

Requests

Reports

Announcements

Settings

Faculty-verified internship records

R.V.R & J.C College of Engineering

Faculty Portal · Department of CSE · Chowdavaram, Guntur, Andhra Pradesh

KS

Dr. K. Srinivasa Rao

Internship Coordinator

Post College Opportunity

Publish an internship and automatically notify every eligible student.

Opportunity details

Company

e.g. Infosys

Duration

8 weeks

Stipend (₹ / month)

15000

Required skills

Java, SQL, Spring Boot

Description

Responsibilities, selection process...

Application link

https://

Role

e.g. Software Engineering Intern

Mode

On-site

Application deadline

dd-----yyyy

Eligible students

221

auto-identified from student profiles

CSE, IT · Year 3 · CGPA 7+

Publish & notify students

Save as draft

Figure 6: Faculty Portal - Reports & Analytics

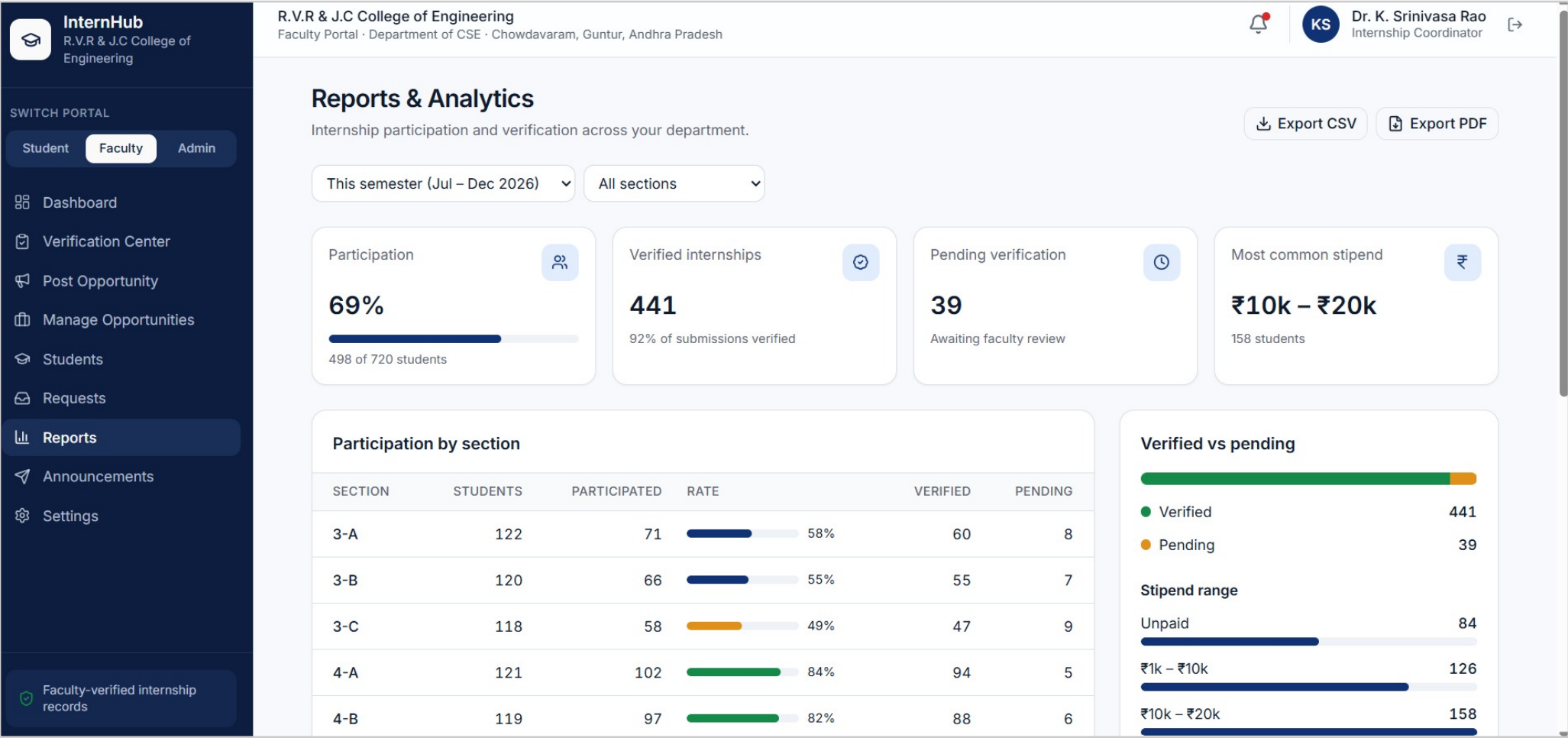


Figure 7: Admin Portal - Platform Overview

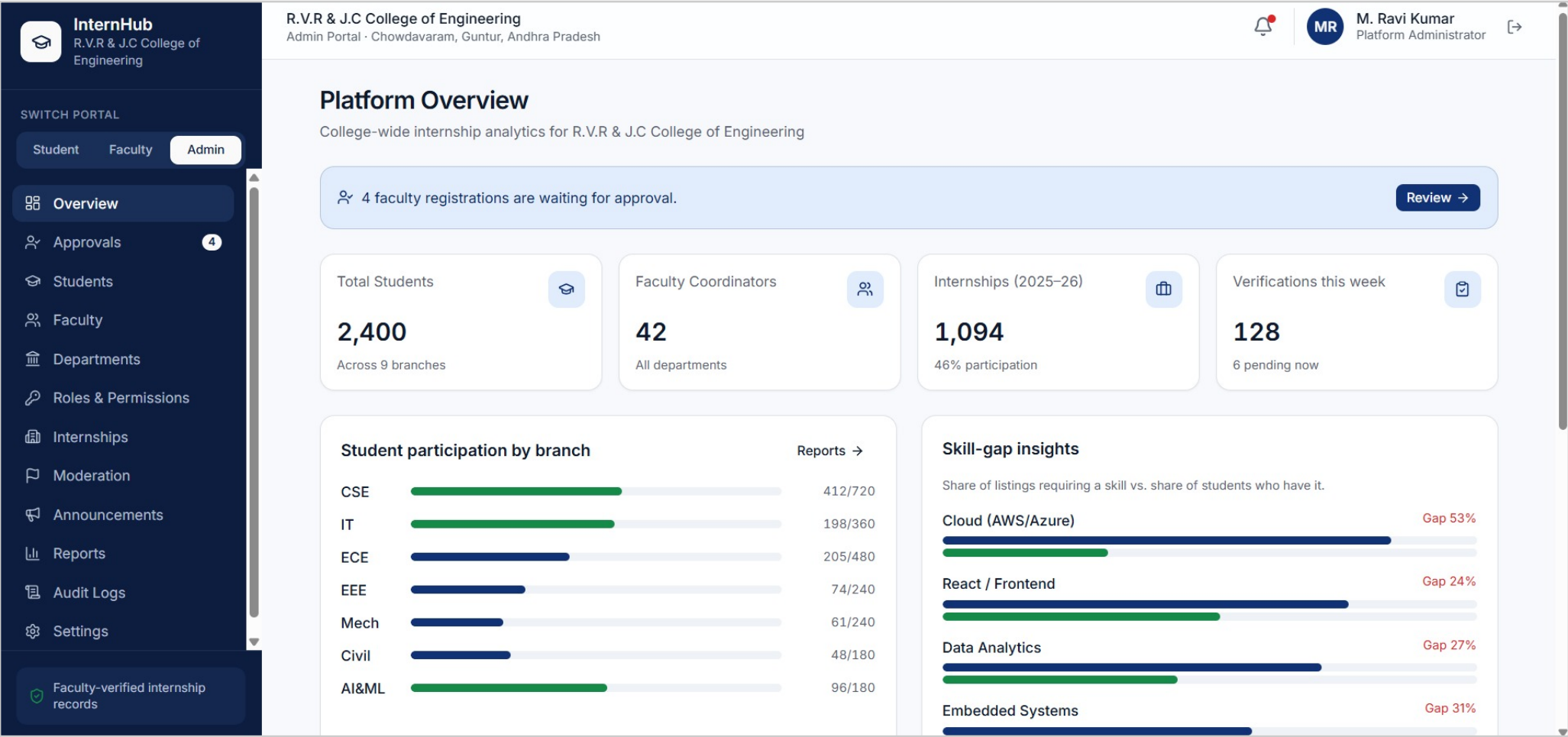


Figure 8: Admin Portal - Approvals

InternHub

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SWITCH PORTAL

Student

Faculty

Admin

Overview

Approvals4

Students

Faculty

Departments

Roles & Permissions

Internships

Moderation

Announcements

Reports

Audit Logs

Settings

Faculty-verified internship records

R.V.R & J.C College of Engineering

Admin Portal · Chowdavaram, Guntur, Andhra Pradesh

MR

M. Ravi Kumar

Platform Administrator

Approvals

New faculty registrations waiting for review

Pending4

Awaiting a decision

Correction requested0

Waiting on faculty

Approved1

Added to Faculty

Rejected1

With a recorded reason

Pending4

Approved1

Rejected1

All6

Faculty	Employee ID	Department	Designation	College email	Submitted	Status			
Dr. R. Hari Prasad	RVR1231	CSE	Assistant Professor	rhprasad@rvrjc.ac.in	07 Oct 2026	Pending	✓ Approve	Request correction	✗ Reject
Mrs. K. Anitha	RVR1236	IT	Assistant Professor	kanitha@rvrjc.ac.in	07 Oct 2026	Pending	✓ Approve	Request correction	✗ Reject
Dr. B. Suresh Babu	RVR1102	Civil	Associate Professor	bsuresh@rvrjc.ac.in	06 Oct 2026	Pending	✓ Approve	Request correction	✗ Reject
Mr. P. Naveen	RVR1240	AI&ML	Assistant Professor	pnaveen.ai@gmail.com Not a college domain	05 Oct 2026	Pending	✓ Approve	Request correction	✗ Reject

Figure 9: Admin Portal - Settings



InternHub
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SWITCH PORTAL

Student

Faculty

Admin

 Overview

 Approvals 4

 Students

 Faculty

 Departments

 Roles & Permissions

 Internships

 Moderation

 Announcements

 Reports

 Audit Logs

 Settings

 Faculty-verified internship records

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M. Ravi Kumar
Platform Administrator



Settings

Platform configuration

General

Verification & SLA

Files & privacy

Notifications

AI & Learning

Support inbox 2

Platform settings

Institution name

R.V.R & J.C College of Engineering

Academic year

2026-27

Verification SLA (days)

3

Max file size (MB)

5

Save settings

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